

Class Diagrams

Package overview

```
classDiagram
    direction TB
    class AuthenticationController
    class AuthenticationService
    class JwtService
    class JwtAuthenticationFilter
    class CustomUserDetailsService
    class BankUserDetails

    AuthenticationController --> AuthenticationService
    AuthenticationService --> JwtService
    AuthenticationService --> CustomUserDetailsService
    JwtAuthenticationFilter --> JwtService
    JwtAuthenticationFilter --> CustomUserDetailsService
    CustomUserDetailsService --> BankUserDetails
```

Account domain

```
classDiagram
    direction TB
    class AccountController {
        +openAccount()
        +searchAccounts()
        +getAccountsByCustomer()
        +updateAccountStatus()
        +deposit()
        +withdraw()
        +transfer()
        +getTransactions()
    }
    class AccountService
    class AccountServiceImpl {
        +openAccount()
        +deposit()
        +withdraw()
        +transfer()
        +searchAccounts()
        +updateAccountStatus()
    }
    class AccountRepository
    class Account
    class AccountNumberGenerator
    class AccountSpecification
    class AccountPolicyController {
        +createPolicy()
        +getActivePolicy()
    }
    class AccountPolicyService
    class AccountPolicyServiceImpl
    class AccountPolicyRepository
    class AccountPolicy
    class Customer
    class TransactionService
    class NotificationEventPublisher

    AccountController --> AccountService
    AccountService <|.. AccountServiceImpl
    AccountServiceImpl --> AccountRepository
    AccountServiceImpl --> AccountPolicyRepository
    AccountServiceImpl --> AccountNumberGenerator
    AccountServiceImpl --> Customer
    AccountServiceImpl --> TransactionService
    AccountServiceImpl --> NotificationEventPublisher
    AccountRepository --> Account
    AccountSpecification ..> Account
    Account --> Customer : customer_id
    AccountPolicyController --> AccountPolicyService
    AccountPolicyService <|.. AccountPolicyServiceImpl
    AccountPolicyServiceImpl --> AccountPolicyRepository
    AccountPolicyRepository --> AccountPolicy
```

Notification (Kafka → email)

```
classDiagram
    direction TB
    class NotificationEventPublisher
    class NotificationEventConsumer
    class BankActionEvent
    class NotificationEmailComposer
    class NotificationMailService
    class SmtplibNotificationMailService
    class SendGridNotificationMailService

    NotificationEventPublisher --> BankActionEvent
    NotificationEventConsumer --> NotificationEmailComposer
    NotificationEventConsumer --> NotificationMailService
    NotificationMailService <|.. SmtplibNotificationMailService
    NotificationMailService <|.. SendGridNotificationMailService
```

Customer domain

```
classDiagram
    direction TB
    class CustomerController
    class CustomerService
    class CustomerServiceImpl {
        +createCustomer()
        +updateCustomer()
        +searchCustomers()
        +deleteCustomer()
    }
    class CustomerRepository
    class Customer
    class AccountService

    CustomerController --> CustomerService
    CustomerService <|.. CustomerServiceImpl
    CustomerServiceImpl --> CustomerRepository
    CustomerServiceImpl --> AccountService
    CustomerRepository --> Customer
```

User / role domain

```
classDiagram
    direction TB
    class UserController
    class UserService
    class User
    class Role
    class UserRole
    class UserRepository
    class RoleRepository
    class UserRoleRepository

    UserController --> UserService
    UserService --> UserRepository
    UserService --> UserRoleRepository
    UserService --> RoleRepository
    User --> UserRole
    Role --> UserRole
```

Shared

```
classDiagram
    direction TB
    class AuditableEntity {
        createdAt
        updatedAt
        createdBy
        updatedBy
        version
    }
    class AccountPolicy
    class User
    class Role
    class UserRole
    class SecurityConfig
    class GlobalExceptionHandler

    AuditableEntity <|-- AccountPolicy
    AuditableEntity <|-- User
    AuditableEntity <|-- Role
    AuditableEntity <|-- UserRole
```

Frontend feature map (logical)

```
flowchart LR
    Login --> AuthService
    Dashboard --> DashboardService
    Customers --> CustomerService
    Accounts --> AccountService
    OpeningDeposit --> AccountPolicyService
    Employees --> UserService
    AuthService --> API
    CustomerService --> API
    AccountService --> API
    AccountPolicyService --> API
    UserService --> API
    DashboardService --> API
```